

Question	Answer	Marks
6(a)(i)	for maximum intensity, waves must be in phase (at detection)	B1
	(phase difference at source means) waves are not in phase / have a 90° phase difference <u>at O</u>	B1
6(a)(ii)	point B labelled on the line PQ below O	M1
	distance OB = distance OA	A1
6(b)(i)	$v = f\lambda$	C1
	$\lambda = (3.00 \times 10^8) / (2.5 \times 10^{10}) = 0.012 \text{ m}$	A1
6(b)(ii)	$\Delta x = 0.012 / 4$ $= 0.0030 \text{ m}$	A1
6(b)(iii)	$\lambda = ax / D$	C1
	distance $= \lambda D / a = 0.012 \times 2.3 / 0.18$ $= 0.15 \text{ m}$	A1

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